BECAMEX BUSINESS SCHOOL



FINAL REPORT

TOPIC: Analysis of Supply Chain Problems and Solutions at Nike

Course: BUS 359 - Operations and Supply Chain Management

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1. Company Overview

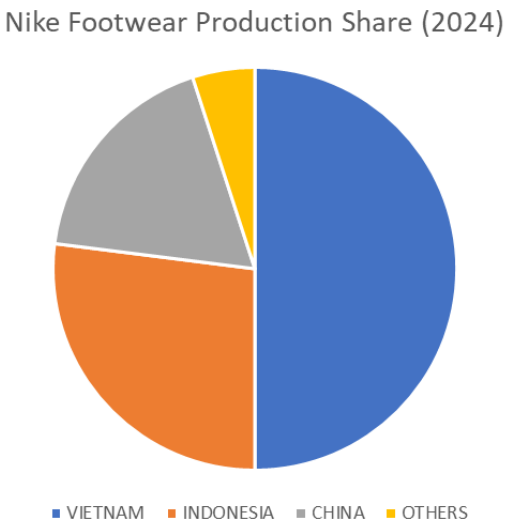
1.1. Background of Nike

Nike, Inc. is known as the world sportswear leader in designing, developing, marketing, and distributing of authentic athletic footwear, apparel, equipment, and accessories. The corporate journey began in January 1964 when the company was founded as Blue Ribbon Sports by University of Oregon track athlete Phil Knight and his coach, Bill Bowerman. Initially operating as a distributor for Japanese shoemaker Onitsuka Tiger, the company quickly realized the potential for the exclusive designs. By 1971, the company began rebranding and changed its name to Nike, Inc., inspired by the Greek goddess of victory. The company's center of operation is located in Beaverton, Oregon, USA. Beside selling sporting goods that the company offers, they also serve consumer services and experiences such as sport events and activations, fitness apps or digital services in retail stores. (Yahoo Finance, n.d.)

1.2. Nike's Global Operations

Nike operates a highly synchronized international network, actively commercializing products and managing intellectual property rights in over 170 countries. A critical element of this global outreach is the brand's physical retail footprint; according to verified tracking data from Statista, Nike operates around 1,048 company-owned retail stores worldwide, including an optimal mix of Nike Factory Stores, premium flagship brands, and Converse retail outlets. (Statista, 2024).

Operationally, Nike sustains this massive supply chain through an asset-light strategy, delegating 100% of its finished goods production to outsourced independent contract manufacturers (Feklist, 2024). By owning no physical manufacturing facilities, the corporation drastically reduces capital expenditure risks and maintains the flexibility to scale production according to fluctuating global demand. These manufacturing facilities are primarily concentrated within specialized industrial corridors in Asia. According to Nike's official regulatory disclosures, Vietnam is a leading manufacturing hub., producing 50% of all Nike footwear, followed by Indonesia at 27% and China at 18%, while regional neighbors make up the remaining 5% of global output (NIKE, Inc., 2024). Apparel manufacturing follows a similarly optimized distribution model centered primarily in Vietnam (28%), China (16%), and Cambodia (15%) (NIKE, Inc., 2024).



1.3. Nike's Business Strategy

Nike's strategy focuses on four key priorities to navigate market fluctuations and maintain brand value (Feklist, 2024). First, performance-oriented innovation is achieved through investment in technology and sustainable engineering. Second, Nike leverages its global leadership to collaborate with top athletes, sponsor clubs, and convey emotionally resonant messages (Vicêncio, 2024). Third, Nike delivers flexibility and cost efficiency in its supply chain through outsourcing (Feklist, 2024). Finally, Nike creates a unified ecosystem by connecting leading retailers both online and offline, such as through the Nike Sneakers app.

2. Nike Supply Chain Overview

2.1. Material Suppliers

Nike uses a variety of raw materials to manufacture its final products and there are six major materials, including polyester, rubber, EVA Foam, cotton, synthetic leather and leather.

Firstly, 19% of the polyester that Nike uses is recycled polyester. This is also the most important ingredient and often produced from recycled plastic bottles to reduce environmental impact. According to the report of the Ochio State University, Nike contributed to recycle approximately 82 million plastic water bottles which were then used to produce polyester. In 2010, all World Cup jerseys were made by Nike, and each one is made from 100% recycled polyester, using eight plastic bottles. China and India are the world's major producers of polyester. In addition, Nike and Syre - Swedish enterprise collaborate long-term on the supply

of recycled polyester from textile waste in 2025. Accordingly, Syre will become a strategic supplier for Nike in expanding the use of recycled polyester. (Mekong ASEAN, 2025)

The next main material Nike uses is rubber with 70% of Nike's footwear is made from environmentally friendly rubber. This is usually used for manufacturing shoe soles because rubber is also known as the most durable materials. Nowadays, some countries in Asia are leading the top of producing rubbers such as Thailand, Indonesia and Myanmar. Although the fluctuation of oil prices can affect synthetic rubber cost, Nike uses plant-based materials that helps reduce this dependence.

Nike also uses organic cotton – a type of cotton grown and produced without the use of pesticides or chemical fertilizers. Therefore, organic cotton has a softer, smoother texture and also possesses high moisture-wicking properties, providing a cooler feel (Dugarco, n.d.). Nike sources most of its cotton from suppliers in China, India, Türkiye, and the United States. However, organic cotton prices have been relatively volatile and have increased significantly. According to the U.S. Bureau of Labor Statistics, the prices of cotton fluctuated and usually increased around 12% from 2021 to 2022.

For leather, Nike collaborates with China, Vietnam, Indonesia and Taiwan that the company always prioritizes the usage of leather that is certified by the Leather Working Group (LWG). For instance, Nike collaborates with Kuo Yuen Tannery which is an LWG Gold-certified leather manufacturer operating in China and Vietnam. This sourcing strategy helps Nike maintain consistent product quality while improving the sustainability of its supply chain (Kuo Yuen Tannery, n.d.).

Nike primarily imports synthetic leather from China and Taiwan. These are also the two leading suppliers of PU leather in the world. Although the supply of synthetic leather is more stable than natural leather, because it is made from petroleum-based materials, changes in global oil prices can directly affect the cost of the synthetic leather that Nike uses in footwear production (Fibre2Fashion, n.d.).

Finally, there's EVA Foam, a material widely used by Nike in shoe sole production to minimize pressure on the feet and provide cushioning. Because it's made from ethylene and vinyl acetate, this means it's highly durable and water-resistant. Most EVA Foam is supplied by China due to its strong industrialization and low costs.

2.2. Procurement Operations of Nike

2.2.1. Virtual Supply Chain Model

Nike has applied a "virtual supply chain" which means the company does not directly build or has its own factories, instead they find outsourcing to complete the final product and then the company just has a duty to monitor progress and product quality. By completely outsourcing, production costs are reduced, and Nike can more easily find alternative suppliers. Furthermore, Nike focuses more on product design, research and development, marketing, and brand management. (SlideShare, 2015).

2.2.2. Outsourcing Strategy

In recent years, some countries in Asia like Vietnam, China and Indonesia have over 103 strategic suppliers and account to manufacture 80% of Nike's products. This reflects the effective strategies of Nike to approach the goals that help Nike reduce operating costs because labor and production costs are lower and affordable compared to other locations. Additionally, it is not overly dependent on any one location. Nike has more ability to become flexible in any emergency situation (ADEC Innovations, 2023).

2.2.3. Procurement and Contract Process

Firstly, Nike has a team of design and development responsible for creating ideas for product designs, specifications, and quality standards for new products. Then, Nike sends this shoe design to a factory and suppliers develop prototype products based on Nike's requirements. After testing and approval, Nike will sign a contract with these factories for mass production., ensuring consistency in product quality and pricing.

2.2.4. Supplier and Production Management

Although Nike does not own factories, the company closely monitors the production process through its quality control teams. Moreover, because Nike applied an outsourcing model, meaning the factories will purchase their own raw materials but must still adhere to Nike's list and control, allowing Nike to manage material costs and only pay after production is complete. Simultaneously, Nike will directly monitor the progress and quality of the product during the production process. (Meey Land, n.d.).

2.2.5. Payment and Cost Control

Nike typically pays its manufacturing partners after the production process is complete and the products meet Nike's quality standards. This encourages suppliers to maintain product quality and meet delivery deadlines. The products will then be shipped to Nike's company, from where Nike will handle the distribution and sales process.

2.3. Manufacturing

Nike collaborates with some manufacturing facilities that mostly are located in Asian countries such as Vietnam, China and Indonesia due to their lower cost labor and strong productivity. Among them, Vietnam is the largest production center, accounting for 50% of global footwear and shoes production, followed by China (27%), Indonesia (18%) and Thailand (3%), while the remaining 2% comes from other countries such as Cambodia, India, and Brazil. (Sellers Union China, 2025).

2.4. Distribution and Logistics

2.4.1. Wholesale Channel

Wholesaler is one of the common channels that generate the largest revenue for Nike, accounting for approximately 68% of Nike's total revenue. (Market Realist, 2020). Through partnerships with major global retail chains such as Foot Locker and JD Sports, Nike distributes athletic shoes, apparel, and accessories to various markets. This allows Nike's products to meet the customers need based on their specific personalization

2.4.2. Direct-to-Consumer (DTC) Channel

Nike's direct-to-consumer (DTC) channel has become the company's fastest-growing business segment. By selling their products directly to the customers, Nike can increase profit margins, be able to collect real customer data and then develop products align with the things that customers want. The DTC channel includes Nike's retail stores and sales channels on e-commerce platforms.

With Nike's retail stores, they operate a variety of retail stores worldwide to diversify and cater to the needs of different customers. These include Nike Factory Stores and Nike Inline Stores, that serve different customer segments. These stores help Nike increase brand recognition and provide a more immersive customer experience.

Nike has invested too much money in areas such as its official website, the SNKRS app, and the Nike By You customization service. Customers can shop online and freely access new items that suit their preferences without worrying about time constraints.

Nike especially provides a unique service called Nike By You. It allows customers to customize the shoes they like. To be more specific, they can remake the color, material, and design of their chosen footwear based on their personal preferences. This strategy has helped Nike attract more new customers and also increase customers. In particular, it also contributed to a 35% increase in the company's sales compared to previous years. (Market Realist, 2020).

2.4.3. Logistics and Transportation

Currently, Nike generates a global logistic network to distribute their products from its manufacturing facility to retailers and customers worldwide. Most of the products are primarily transported via a combination of sea and air freight and Nike partners with third-party logistics (3PL) providers such as UPS, FedEx, and Maersk to manage global shipping, warehousing, and distribution. This approach helps Nike improve operational flexibility, reduce costs, and increase its ability to respond more quickly to changes in market demand. (ALS, n.d.).

3. Symptoms

The Nike supply chain has several operational symptoms that have a negative impact on production efficiency, logistics performance and inventory management. Nike heavily relies on outsourcing for production, and its distribution networks are spread across the globe; therefore, disruptions at one stage can impact activities further down the line. Three key symptoms found are delivery delays, supply production capacity limitations and inventory imbalance (Supply Chain Dive, 2021; Sellers Union China, 2024).

3.1. Delivery Delays

Nike's major problem in the global supply chain is late delivery. Nike products use Asian countries, particularly Vietnam, China and Indonesia, as contract manufacturing bases, which results in long transportation routes from these countries to major markets like North America and Europe (Sellers Union China, 2024).

One significant case in point was the length of time it took to ship products from Asia to North America. Transit time rose from about 40 days to almost 80 days, disrupting retailers'

replenishment and seasonal product launches. Sneakersnstuff retailers also had weeks or months of waiting time for Nike products (Supply Chain Dive, 2021).

Operational Causes

There are delivery delays due to inflexibility in production at supplier factories and transportation inefficiency. Since Nike is outsourcing, shipping times are dependent on suppliers delivering the product on time. In times of high demand, factories of the suppliers might be unable to take emergency orders because of their limited production flexibility.

Adverse Operational Impacts

Poor delivery timing has a negative impact on product availability, customer satisfaction and seasonal product launches. Air freight could also be used as a way to buffer against delays, adding to Nike's logistics expenses. In addition, slow delivery times decrease Nike's responsiveness to market demand and could cause customers to turn to competing brands.

3.2. Production Capacity Constraints at Supplier Level

Another major symptom in Nike's supply chain is production capacity shortages at third parties. Nike's outsourcing manufacturing policy helps to decrease the company's costs, but it results in decreased manufacturing control and flexibility.

Nike has a very concentrated manufacturing network: approximately 50% of its footwear production happens in Vietnam, 27% in China and 18% in Indonesia (Sellers Union China, 2024). This focus brings economies of scale but also a single-point-of-failure problem in the event of disruption at key production centres.

Key Capacity Challenges

Production constraints can arise due to a shortage of manpower, machines or production lines in factories during periods of peak demand. Moreover, there may be delays in supplier factories if they need to change from one product model to another or if they need to process urgent orders.

Adverse Operational Impacts

Production constraints by suppliers may cause order delays and hamper logistics processes. Production inflexibility further hampers Nike in meeting the demands of the market with quickness, adding to delays in delivery.

3.3. Inventory Imbalance

The other significant symptom in Nike's supply chain is inventory imbalance, meaning that some markets have insufficient inventory while other markets have surplus inventory. A 65% increase in inventory was reported by Nike in North America due to under-forecasting and delivery delays (Retail Dive, 2022).

The Bullwhip Effect and Demand Distortion

The Bullwhip Effect can be understood in this context: if customers' demand for a product fluctuates slightly, these changes can be magnified upstream in the supply chain (Lee et al., 1997). Some forecasts were wrong and others were delayed, which meant that some areas were short of stock while others were overstocked in Nike’s system.

Adverse Operational Impacts

The imbalance in inventory drives up warehousing expenses, decreases profitability through markdown pricing, and decreases customer satisfaction by reducing product availability. The biggest problem is that overstock and stock shortages coexist, meaning that demand forecasting, inventory planning and regional allocation are not well synchronized.

4. Problem Identification (Ishikawa Analysis)

4.1. Inefficient Production and Logistics Flow

Nike's global supply chain is often delayed due to its reliance on outsourcing to factories located throughout Asia (Vietnam, China, and Indonesia). The long distances involved in transporting goods, combined with the difficulty of exercising control over suppliers' operations, limit Nike's flexibility to meet the demands for product availability in the marketplace. Consequently, logistics costs increase because of the increased cost of using air freight to lessen the time it takes to distribute products.

4.1.1. Ishikawa (Fishbone Summary)

The Ishikawa (Fishbone) analysis shows that the inefficient production and logistics flow of Nike is due to several operational factors. The lack of labor and high turnover of workers in supplier factories reduces production efficiency and increases the possibility of delays from a manpower point of view. On the machines and capacity side, supplier factories have limited production lines and capacity constraints, especially in high demand periods.

In the issue problems are in methods as well. The fixed production planning and the long setups when changing the product models reduce the agility and make prompt delivery impossible. Nike relies heavily on its external suppliers. There are the following supplier based problems: supplier capacities and raw material supply.

On the analysis side, problems also exist concerning the measurements: lack of correspondence between production volume and logistics; delays in information feedback; lack of clarity on logistics operations. Beyond these, external sources like changing consumer demand, uncertainty of the market and traffic problems can influence logistics operations.

The fishbone analysis also shows that the inefficient Nike production-logistics flow results from low supplier flexibility, operational deficiencies and bad coordination in worldwide supply chains.

4.2. Poor Demand-Inventory Synchronization

Nike's supply chain is hampered by uneven stock, resulting in both shortfalls and surpluses across markets. Reasons for this include incorrect demand estimation, poor and lagging communication, and poor communication between inventory control and market allocation. The Bullwhip Effect aggravates these problems, amplifying even slight changes in consumer demand within the supply chain.

As a result of this inventory imbalance, Nike has incurred higher warehousing costs, lost potential sales from markdowns, and lower levels of customer satisfaction due to inconsistent product availability.

4.2.1. Ishikawa (Fishbone Summary)

According to an Ishikawa Analysis, Nike's ineffective demand/inventory synchronization is caused by several operational challenges in its global supply chain. The analysis indicates that

from a personnel standpoint, weak communication and a lack of coordination between regional planning teams inhibit forecasting accuracy. From a methods perspective, Nike's centralized forecasting system and rigid inventory allocation process are not able to quickly adjust to changing local market requirements.

Nike is encountering issues with real-time inventory visibility because they do not have a unified inventory model. Additionally, incorrect predictions and delayed data distribution create issues with their inventory planning. Many challenges come with materials, including receiving insufficient products or being unable to rely on suppliers for deliveries, these all affect maintaing a steady inventory balance.

Nike is facing both excess inventory and shortages at different locations, negatively impacting their operational efficiency, customer satisfaction and overall performance.

5. Key Problem Selection

5.1. Criteria for evaluation

Based on previous analysis, Nike is facing three main problems in its supply chain networks which are delays in delivery, limited production capacity and inventory imbalance. Among these issues, the mismatch between demand and the inventory is considered a key problem as it has a major effect on the overall supply chain network.

According to Lee et al. (1997), false information about demand can lead to the Bullwhip Effect which makes the demand is amplified across the supply chain. For Nike, this inventory imbalance increased warehouse costs and lost sales opportunities.

In contrast, delivery delays and production capacity constraints are not considered as main problems because they are caused mainly by operational activities or planning issues. Delivery is determined by upstream activities such as demand forecasting and inventory (Christopher, 2016). Therefore, solving the problem of poor demand-inventory synchronization can help address both inventory and other operational issues in Nike's supply chain.

5.2. Selected Key Problem

Poor demand-inventory synchronization is a key problem in Nike's supply chain because it leads to both surpluses and shortages of goods. When demand forecasting is inaccurate, Nike

can produce products more than customers are willing to buy. Thus, the company will suffer from surplus inventory, higher warehousing rent cost and markdowns that reduce profit margins. Lack of available products with popular products also lose sales opportunities and customer dissatisfaction.

A notable example occurred when Nike implemented a new demand forecasting system that created inaccurate demand estimates. This system overestimated demand for less popular products like Air Garnett III, whereas underestimated demand for higher demand products like Air Jordan. Therefore, retailers faced excess inventory of slow-moving products and had to heavily discount them, while popular products were out of stock. For instance, Foot Locker was forced to lower the price of Air Garnett to $90 instead of the expected retail price of $140 in order to sell the stock. Production delays further forced Nike to use air freight costing between $4 and $8 per pair instead of ocean freight at approximately $0.75 per pair. Overall, the failure in forecasting lost about 100 million USD revenue, highlighting the significant impact of poor synchronization between demand forecasting and inventory planning (Dolfing, 2022).

6. Alternative Solutions

6.1. AI-Based Demand Forecasting System

AI-Based Demand Forecasting System uses AI to forecast customer demand to increase the accuracy. The forecasting system uses many different data sources such as historical data, regional purchasing behavior, fluctuations in seasonal demand, and promotional activities. Many machine learning models such as Long Short-Term Memory (LSTM) and Extreme Gradient Boosting (XGBoost) help handle a large number of complex datasets while giving predictions. According to Fatima and Salam (2026), adding external factors such as holidays, weather conditions, weekdays, and sales information into machine learning forecasting models improves forecasting accuracy better than traditional models. Their study also found that XGBoost achieved the highest accuracy when these contextual variables were included.

In reality, AI forecasting systems analyze both historical and real-time data to predict future demand in different regions and product categories. This helps learning models quickly adapt to changing customers behaviors and market trends. As a result, this allows companies to respond faster to sudden increases or decreases in demand and improve the inventory performance. A study by DreamzTech (2024) reported that AI helped improve forecast accuracy

from 68% to 94% while reducing stockouts by 42%. In addition, research by Infiria (2024) also found that machine learning forecasting systems with real-time data integration help reduce inventory and stock-related problems. These findings suggest that AI forecasting can greatly improve inventory management performance in large global supply chains.

6.2. Regional Inventory Allocation System

Another alternative solution is the implementation of a regional inventory allocation system. This means Nike changes from centralized inventory structure to a regional one which helps the inventory be stored closer to each market demand. By using this method, each regional hub is responsible for inventory and activities based on specific local demand. According to Wang et al. (2023) who investigated a multi-echelon location inventory problem including regional distribution centers and found that decentralized inventory allocation coordinated replenishment strategies can improve delivery efficiency and reduce supply chains cost. These hubs can improve supply chain systems by holding safety stock and respond more quickly to changes in customers' behaviors. Not only that, regional inventory hubs improve supply chain responsiveness and reduce lead times because the products are positioned closer to end customers. In addition, regional inventory can help Nike respond more effectively to rapid changing fashion trends. They also help reduce the risk of global inventory imbalance and delays caused by long international shipping routes.

6.3. Real-Time Integrated Supply Chain Platform

Real-Time Integrated Supply Chain Platform focuses on improving information and coordination across departments and the entire supply chain networks by connecting suppliers, factories, warehouses, logistics companies, retailers and distribution channels into one digital system. In other words, it works like a digital “control tower” defined as a digital system allowing connection between supply chain data, business data and important events in one place (IBM). By that way, Nike can monitor supply chain activities in real time, share and update information across its network.

Additionally, data analytics tools also support managers to analyze large amounts of numbers and supply chain data, then identify potential problems. According to Ivanov and Dolgui (2020), real-time digital integration improves supply chain coordination and helps companies respond faster to disruptions and operational problems. Not only that, they also help improve inventory visibility and faster decision making. This method also helps sharing information which reduces

communication gap between different partners and departments. This is because Nike relies a lot on outsourced manufacturer and third-party logistic companies, faster and more accurate information sharing is important.

7. Evaluation & Workable Solution

7.1 Evaluation

7.1.1. AI-Based Demand Forecasting System

AI-Based Demand Forecasting System helps improve forecasting accuracy because Nike sells thousands of products globally but preferences, habits of customers are different across countries. Furthermore, external factors including geography, season, sport events and fashion trends change quickly. AI forecasting systems are able to combine these factors to generate more accurate demand predictions. According to Fatima and Salam (2026), machine learning forecasting models integrate factors such as holidays, weekdays and seasonal demand to create results that are much better than traditional methods. The research also showed that the XGBoost forecasting model is the model that achieved the highest prediction accuracy because it can process complex data more efficiently.

However, there are still several challenges. Firstly, the use of AI depends heavily on the quality of the data and the system. Nike operates outsourced involving many suppliers, manufacturers, and logistics partners and when the data from these partners is inaccurate or outdated, forecasting performance may decline. According to Choi et al (2018), low data quality and fragmented information systems are one of the biggest barriers to successful AI implementation in supply chain management. Secondly, implementing AI forecasting systems requires a lot of investment from technology infrastructure, computing systems, employee training to software development. Although the long-term benefits may be much higher, the implementation process is still expensive and complex. Finally, AI forecasting systems do not always provide accurate information in some situations. Sudden fashion trends, social media trends, influencers or unexpected shifts in customer tastes can reduce forecasting reliability even when advanced machine learning models are used. Therefore, decision making from experienced managers and market monitoring are still necessary to support forecasting decisions.

7.1.2. Regional Inventory Allocation System

Regional inventory systems help Nike reduce delays in delivery time and respond faster to changes in customer demand in different regions because demand for clothing products is different across places and depending on fashion trends, sports events, culture, and climate. For example, products related to footfall are popular in Europe during major tournaments, while basketball products are loved in North America. A regional allocation system allows Nike to store and distribute goods based on local demand instead of a centralized warehouse. Secondly, it makes the supply chain more flexible. Nike currently depends on outsourced manufacturing and global transportation networks which can raise risks when demand changes unexpectedly. Regional inventory allocation not only helps reduce dependence on long shipping routes, reducing time but also allows the company to manage inventory more efficiently in different markets.

However, this solution also has some limitations. First, operating a number of regional distribution centers increases warehousing, inventory, and operational costs. According to Christopher (2016), decentralized inventory management systems improve responsiveness but may reduce economies of scale because companies have to maintain inventory in many locations from many countries at the same time. As a result, Nike may face greater challenges in managing in different regions which may make the coordination more complex and require stronger communication between partners. Research by Kaminsky et al, (2008) emphasized that decentralized supply chains require more coordination and information sharing to avoid inventory duplication and inefficient allocation. It also needs strategic planning and close collaboration with logistics partners.

7.1.3. Real-Time Integrated Supply Chain Platform

Real-Time Integrated Supply Chain Platform improves real-time coordination between Nike's partners and departments since it relies heavily on an outsourced global supply chain, any delays and mismatch information between them affect heavily on inventory allocation and delivery. The solution integrates tools such as ERP, cloud computing, IoT, and other analytics to help information to be shared more quickly and efficiently across global operations. For example, if delays happen at Vietnamese factories, managers can immediately recognize and adjust transportation schedules or decisions before the disruption in one place affects the whole supply chain network. According to Sanders et al. (2019), companies that use integrated digital

systems supporting supply chains often achieve better results, coordination and faster responses because managers can access updated and accurate data.

The main challenges of this solution include high cost and complexity. To be specific, digital platforms require large amounts of investment costs. So, Nike has to invest heavily in computing infrastructure, ERP systems, cybersecurity, software, employee training, and data management. Research by Ivanov and Dolgui (2020) proved that digital transformation projects in global supply chains are often expensive and complex because they involve integrating data from a lot of organizations and regions. Furthermore, the implementation can be complicated because Nike relies mainly on many external providers and third-party logistics companies. Different suppliers and partners use different information systems, making it difficult and complex to integrate into one system.

7.2. Final Workable Solution

The solution that is the most suitable for Nike is a Hybrid Regional Inventory Allocation System with AI-Based Demand Forecasting. Because it takes advantage of the strengths of both forecasting and regional inventory. It helps in saving cost and is easy to implement. It also helps in analyzing elements such as historical data, seasonal demand, preferences, purchasing power, and market trends to predict future demand. The forecasting system helps determine which products experience high or low demand in different regions so that Nike can allocate inventory more effectively across regional distribution hubs.

For example, during sports events such as the FIFA World Cup, products such as jerseys, football shoes, and training equipment are often in high demand. AI forecasting systems can collect and analyze data from previous events, how football is popular across regions, and customer reactions so that Nike can allocate inventory more efficiently, reduce stockout and delivery delays during peak demand.

In addition, Nike can cooperate with third-party logistics (3PL) such as DHL and FedEx while still avoiding building all warehouses by itself because they have existing large warehouses, transportation systems, and distribution infrastructure. By that way, Nike can use regional inventory hubs while saving a lot of money on land, labor, and facilities. A study by ForthMatch (2024) found that storing products closer to customers through regional 3PL can reduce last-mile delivery costs by about 20–30% and improve delivery speed. Nike can also store common products in regional warehouses, while uncommon products are kept in centralized

warehouses. This helps improve delivery time for popular items while reducing unnecessary costs for low demand ones.

8. Conclusion

Although Nike still has disadvantages in managing inventory such as delays, capacity limitations, and inventory imbalances, analysis has pointed to a core problem: weak coordination in demand forecasting, inventory planning, and the supply chain is the cause of these issues. However, a hybrid approach of regional inventory allocation and AI forecasting is the most suitable. By improving the accuracy and distribution of inventory, Nike can minimize inefficiencies, enhance customer satisfaction, and maintain its competitiveness in the international market.

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